TU Darmstadt M.Sc. AI and Machine Learning: Academic Eligibility Checklist

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#	Reference module	My matching courses (credit hours, grade %)	Credit hours	ECTS
1	**Functional and Object-oriented Programming Concepts**	CS 103 Programming I (3 cr, 85) CS 104 Programming II (3 cr, 87) CS 209 Advanced Programming (3 cr, 91)	9	15.4
2	**Algorithms and Data Structures**	CS 201 Data Structures and Algorithms (3 cr, 83) CS 205 Algorithms Design (3 cr, 87)	6	10.3
3	**Digital Design**	CS 203 Logic Circuit Design (3 cr, 96)	3	5.1
4	**Computer Organisation**	CS 204 Computer Architecture (3 cr, 81) CS 309 Embedded Systems (3 cr, 98)	6	10.3
5	**Parallel Programming**	*No matching course*		
6	**Operating Systems**	CS 206 Operating System (3 cr, 88) CS 304 System Programming (3 cr, 94)	6	10.3
7	**Introduction to Compiler Construction**	CS 404 Compiler and Language Theory (3 cr, 97)	3	5.1
8	**Automata, Formal Languages and Decidability**	CS 307 Theory of Computation (3 cr, 80)	3	5.1
9	**Propositional Logic and Predicate Logic**	CS 101 Discrete Structures (3 cr, 89)	3	5.1
10	**Formal Methods in Software Design**	*No matching course*		
11	**Computer Networks and Distributed Systems**	CS 301 Computer Networks (3 cr, 80) CS 403 Advanced Computer Networks (3 cr, 92) CS 405 Distributed Systems and Cloud Computing (3 cr, see transcript)	9	15.4
12	**Computer Security**	SS 102 Introduction to Computer Security (3 cr, 93)	3	5.1
13	**Information Management**	CS 202 Database Systems (3 cr, 87) DS 411 Healthcare Database Systems (3 cr, 95)	6	10.3
14	**Software Engineering**	CS 106 Software Engineering (3 cr, 80) CS 311 Software Quality Assurance and Testing (3 cr, 95)	6	10.3
15	**Modeling, Specification and Semantics**	*No matching course*		
16	**Visual Computing**	CS 308 Computer Graphics (3 cr, 95)	3	5.1
17	**Introduction to Artificial Intelligence**	AI 102 Introduction to Artificial Intelligence (3 cr, 85) AI 201 Intelligent Systems (3 cr, 89) AI 301 Machine Learning (3 cr, 94)	9	15.4
18	**Probabilistic Methods in Computer Science**	MATH 201 Probability and Statistics (2 cr, 87)	2	3.4
19	**Scientific Computing**	MATH 301 Numerical Computations (2 cr, 97) CS 207 System Simulation and Modelling (3 cr, 85)	5	8.6
	Total (minimum required: 60 ECTS)		**82**	**140.6**

ECTS conversion: my degree has 140 credit hours over 4 years (8 semesters). Using 60 ECTS per year, 140 credit hours = 240 ECTS, so 1 credit hour = 1.71 ECTS (3 credit hours = 5.1 ECTS, 2 credit hours = 3.4 ECTS).
Course descriptions: the official Pharos course descriptions are attached with the application.